Kunal Garg

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Ph.D. candidate in Economics with a strong quantitative and computational background, experienced in large-scale data analysis, structural and reduced-form methods, numerical optimization, and causal inference, with strong Python proficiency and exposure to machine learning techniques.

Education

Ph.D. in Economics — Texas A&M University, 2022 -- 2027 (expected)

M.Sc. in Quantitative Economics — Indian Statistical Institute (Delhi), 2018 -- 2020

B.A. in Economics — University of Delhi, 2015 -- 2018

Professional experience

Research Assistant — Department of Economics, Texas A&M University, 2023/06 -- 2025/08

- Used Google and Yelp APIs to create and merge proprietary datasets with public data, applying clustering, geocoding, and string distance algorithms for matching.
- Implemented a structural model of firm entry under strategic interaction using Python, solving a multi-stage decision problem via backward induction, contraction mappings, and numerical optimization on simulated data.
- Conducted extensive empirical analysis across multiple datasets, including firm-level flood exposure, industry-level unemployment insurance payments, and sales tax data, to study firm closures, re-entry patterns, and broader economic effects of disaster relief.

Research Associate — CAFRAL, Reserve Bank of India, 2021/05 -- 2022/04

- Explored CNN-based modeling on satellite imagery, training models and examining whether learned features could predict socioeconomic indicators via ridge regression.
- Analyzed matched firm–bank panel data using triple-difference DiD regressions with firm fixed effects, interpreting high-dimensional interaction effects to evaluate policy impacts on bank lending behavior.
- Performed exploratory analysis of India's NBFC sector using proprietary RBI and CMIE data following major failures.

Research Associate — Graduate School of Business, Krea University , 2021/03 -- 2021/04

- Supported design and implementation of a randomized controlled trial (RCT) in India, contributing to treatment–control assignment and identification strategy to ensure clean causal inference.
- Coordinated field logistics across nine districts, including stakeholder approvals and seminar execution, to support experimental integrity and data collection.

Teaching Fellow — Department of Economics, Ashoka University , 2020/08 -- 2021/05

- Led tutorial sessions and recitations for undergraduate and graduate courses in economics and quantitative methods.
- Supported instruction through grading, exam and problem-set preparation, and student support in mathematics, statistics, and Python programming for economics.

Research Experience

1. **"Understanding the Flood Insurance Gap: Factors Influencing Adoption and Renewal Behavior"** (Ongoing)
   - I construct a novel large-scale panel of NFIP flood insurance policies from 2009–2019 (≈70 million observations), enabling tracking of individual policyholders over time. I document heterogeneous renewal and cancellation behavior by flood-risk exposure and policy tenure, and develop a Bayesian belief-updating framework with attention to study repeated insurance decisions.
2. **"Corporate Board Management and Greenhouse Gas Emissions"** (Ongoing; With coauthors)
   - We examine how mandatory environmental disclosure affects board composition by exploiting a regulatory reporting shock, constructing merged administrative and governance datasets and applying Synthetic DiD methods, while exploring potential implications for director compensation.

Skills

Python, MATLAB, R, Stata; experience with prediction and classification, machine learning methods, and convolutional neural networks (CNNs). Familiar with SQL; experience working with large administrative and geospatial datasets; LaTeX.

Languages: English (fluent), Hindi (native).

Interests: Variant Sudoku, chess (amateur), playing sports, and reading detective fiction.